MILITARY SPECIFICATION

ANESTHESIA APPARATUS, GAS, NITROUS OXIDE

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable anesthesia apparatus for use in the field in the administration of nitrous oxide, oxygen, and liquid anesthetics for inhalation anesthesia.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-C-77	-	Casters, Rigid and Swivel (Institutional Duty).
TT-E-527	-	Enamel, Alkyd, Lustreless.
TT-P-1757	-	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.

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MIL-T-5542	-	Thread Compound, Anti-Seize Air and Oxygen Systems.
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595 - Colors.
FED. TEST METHOD STD. NO. 601 - Rubber: Sampling and Testing.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-810 - Environmental Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24441 - Anesthesia Apparatus, Gas, Nitrous Oxide, Oxygen and Volatile Liquid Anesthetics, Portable, 4 Cylinder Capacity.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 56A - Standard for the Use of Inhalation Anesthetics.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 247 - Specification for Aluminum-Alloy Die and Hand Forgings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B57.1	-	Compressed Gas Cylinder Valve Outlet and Inlet Connections.
ANSI Z79.2	-	Tracheal Tube Connectors and Adapters.
ANSI Z79.8	-	Minimum Performance and Safety Requirements for Components and Systems of Continuous-Flow Anesthesia Machines for Human Use.
ANSI Z79.11	-	Anesthetic Equipment - Scavenging Systems for Excess Anesthetic Gases.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)

CGA Pamphlet M-1	-	Standard for 22MM Anesthesia Breathing Circuit Connectors.

(Application for copies should be addressed to the Compressed Gas Association, Inc., 500 Fifth Avenue, New York, NY 10036.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fire and casualty hazards. The rubber components and casters shall comply with the fire and casualty hazards provisions of the applicable standards for operation in hazardous locations, Class I, Group C, and as prescribed by the National Fire Protection Association (NFPA) No. 56A, and be an approved type for the purpose intended, as determined by a nationally recognized testing agency, adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp, or label to the article, or issuance of a certificate. The stamp, label, seal, or certificate of the Underwriters' Laboratories, Inc. (UL), or other testing laboratory approved by the contracting agency shall be considered acceptable evidence of each approval. However, in addition to the above, the contracting agency reserves the right to require the contractor to submit a laboratory test report which shall include test results and methods used to determine conformance. Compliance with the above requirements regarding fire and casualty hazards does not absolve the

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contractor from complete compliance with all other requirements of this specification.

3.2 Storage and transportation. The entire apparatus shall be capable of safe storage and transportation without impairment of its capabilities from the effects of extreme environmental conditions. The apparatus shall be subjected to the following environmental test methods of MIL-STD-810:

a. High temperature storage. High temperature test as specified in Method 501, Procedure I, except that the temperature shall be 155°F with absolute humidity of 13 grains per cubic foot, for periods of up to 4 hours.

b. Low temperature storage. Low temperature test as specified in Method 502, using a temperature of -65°F for 12 hours.

c. Air transit conditions. Altitude test as specified in Method 500, Procedure I, except that the internal pressure shall be reduced to 5.00 inches of mercury (40,000 feet) instead of 3.44 inches of mercury and, after one hour, raised to 254 millimeters (mm) of mercury (10.0 inches of mercury or 26,000 feet above sea level).

3.2.1 Interchangeability of repair parts. The individual parts, or assembly of parts (where applicable), of each anesthesia apparatus supplied shall be interchangeable with the respective parts, or assemblies of parts, of every other apparatus of the same manufacturer and model, without adverse effect on the functioning of the end item.

3.3 Material.

3.3.1 Metal components. Metallic components shall be fabricated from corrosion-resistant materials. All external brass components shall be nickel, or chromium over nickel plated. The operative head, vaporizer, and absorber base shall be nickel plated.

3.3.2 Rubber. Rubber components shall be of material specified herein. Where material is not specified, the rubber shall be natural or synthetic, or a combination of the two. All rubber components shall be conductive (except internal components such as "O"-ring seals and gaskets) and have a maximum measured resistance of 1 megohm when tested as specified in 4.4.1. Rubber parts shall withstand the accelerated aging test specified in Method 7221 of FED. TEST METHOD STD. NO. 601, without becoming tacky or brittle or decreasing in tensile strength by more than 25 percent. Test specified in FED. TEST METHOD STD. NO. 601 need not be performed on components other than the rebreathing bag provided a certificate is supplied stating that all other rubber components not tested are fabricated from the same batch of rubber material as the rebreathing bag.

3.3.3 Other material. Other material shall be of high commercial quality throughout, and suitable in every respect to serve the purpose for which it is intended.

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3.4 General description. Shall be an anesthesia apparatus which shall consist, essentially, of an operative head, absorber, vaporizer, pressure regulators, flexible hoses with end connections, rubber goods, "Y"-connector, and various accessories all fitted within a carrying case that is an integral and functional part of the unit. The anesthesia apparatus shall be self-contained and shall be suitable for use in the field in administering oxygen, nitrous oxide, and volatile liquid anesthetics, with acceptable professional versatility. The apparatus shall permit employment of closed, semi-closed, and open circuit anesthesia techniques.

3.5 Construction, style, and design. Construction, style, and design of the anesthesia apparatus shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24441. The drawing defines the case, the mount, and various accessories. The operative head assembly shall be shown as a pictorial representation only. Hereafter, when "drawing" is designated it shall refer to DPSC Drawing 24441.

3.5.1 Circuitry. The circuitry shall be in accordance with the schematic flow diagram as shown on the drawing and as hereafter indicated. The circuitry shall consist of gas dispensing circuits and a breathing circuit.

3.5.1.1 Gas dispensing circuits. The oxygen and nitrous oxide circuits shall each be provided with two regulators for use with flush type (medical post) cylinder valves and one regulator adapter for screw connection valves. One long and one short hose each shall be provided for oxygen and nitrous oxide. There shall be one needle valve and flowmeter, each for oxygen and nitrous oxide, and one needle valve and two flowmeters for oxygen supply to the vaporizer.

3.5.1.2 Breathing circuit. As arranged for closed or semi-closed technique, circuit shall include, in the following sequence, a "Y"-connector, a corrugated breathing or supply tube, an exhalation check valve, a rebreathing bag, a spring-loaded (non-calibrated) adjustable relief valve, a carbon dioxide absorber, a breathing pressure gage, a spring-loaded non-adjustable relief valve, an inhalation check valve, and another corrugated breathing or supply tube. The accessory items required to convert to the pediatric open circuit techniques shall include a one liter bag, a short corrugated breathing tube, a supply hose, a Tee-connector, and an adapter. One each medium adult and large adult anatomical masks, and one each newborn, pediatric, and child size Rendell-Baker-Soucek masks shall be provided for the connection of the patient to the breathing circuit.

3.6 Operative head assembly. The operative head assembly shall include gas inlet connections, needle valves, flowmeters, inhalation and exhalation check valves, an oxygen flush valve, spring-loaded, adjustable pressure relief valve, circuit pressure gage, an oxygen-powered shut-off valve for nitrous oxide, an absorber unit, a vaporizer, a rebreathing bag connection, a spring-loaded nonadjustable relief valve, supply tubing connections, and two utility tray mounting studs. The operative head shall be sufficiently gas-tight under both vaporizer valve positions when tested as specified in 4.4.2.1. The top manifold of the head assembly shall serve as a handle to permit lifting of the head from the case when the unit is erected for use.

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3.6.1 Operative head. The operative head body shall be fabricated from forged brass and shall be nickel plated. The outputs of the oxygen and nitrous oxide flowmeters shall be connected to the operative head forging. The output of the vaporizer flowmeters shall be externally delivered to the vaporizer.

3.6.1.1 Gas inlet connections. The oxygen and nitrous oxide inlet connections shall mate with Schrader No. 4951 and No. 4826A automatic check valves, or equal, respectively. The operative head shall be permanently and legibly stamped with the symbols "O_2" and "N_2O", adjacent to the inlet connections for oxygen and nitrous oxide respectively. The connections shall be identified in a similar manner. The gas inlet fittings shall be secured to the head in such a manner that a minimum torque of 150 inch pounds shall be required to remove a fitting from the head. The inlet connection shall mate noninterchangeably with the outlet connections of the corresponding flexible hose assemblies and shall be readily accessible on the rear of the operative head when the apparatus is erected to its normal operating position.

3.6.2 Needle valves. Needle valves shall be associated with their corresponding flowmeters and identified by color (white for oxygen, blue for nitrous oxide, yellow for vaporizer) and symbol with the gas controlled. All parts of the needle valves shall be replaceable. An adjustable stop shall be provided to compensate for wear and prevent excessive sealing stress. Needle shall be fabricated from corrosion-resisting steel and shall mate with a valve seat fabricated from silver alloy having a composition of 99.8 percent silver and 0.2 percent copper. Each needle valve shall be sufficiently sensitive so that with an input oxygen pressure of 40 pounds per square inch gage (psig) the change in flow rate from 1 liter per minute (LPM) to 6 LPM requires not less than 1/3 turn of the needle valve knob.

3.6.3 Flowmeters. Oxygen flowmeter shall be calibrated at 200, 400, 600, and 800 cc per minute and 1, 2, 3, 4, 5, 6, and 7 LPM. Nitrous oxide flowmeter shall be calibrated at 200, 400, 600, and 800 cc per minute and 1, 2, 3, 4, 5, 6, 7, and 8 LPM. The low flow vaporizer flowmeter shall be calibrated at 20, 40, 60, 80, 100, 120, 140, 160, and 180 cc per minute, and the high flow vaporizer flowmeter at 100, 200, 300, 400, 500, 600, 700, 800, 900, and 1000 cc per minute. The calibrated scale range of flowmeters shall be no less than 2-1/2 inches in length. Flowmeter scale for oxygen shall be white background with green characters; for nitrous oxide, scale shall be blue background with white characters; and for vaporizer shall be yellow background with black characters. The maximum error when the flowmeter is tested as specified in 4.4.4 shall not exceed + 20% at flow rates from 0 to 1000 cc per minute of the indicated scale reading and + 10% at flows greater than 1000 cc per minute of the indicated scale reading when tested as specified in 4.4.4. The indicators of the flowmeters shall not pulsate or jump during normal operation with spontaneous respiration. The flowmeter scales shall provide contrasting colors to the floats.

3.6.3.1 Flowmeter shield. A clear flowmeter shield shall be assembled to the unit, in front of the flowmeters. The shield shall protect the entire height and width of the assembled flowmeters. Shield shall be fabricated from a durable plastic that shall not obstruct viewing the flowmeter scales.

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3.6.4 Flowmeter manifold. The flowmeter manifold shall be fabricated from forged aluminum in accordance with ASTM B 247 for alloy 2014-T6. The flowmeter manifold shall bear permanent and legible markings indicating "VAPORIZER FLOW", directions for reading the flowmeter floats (dead center on top or bottom, etc., as applicable), "MODEL 885A", serial number, and manufacturer's name or registered trademark.

3.6.5 Oxygen flush valve. The oxygen flush valve shall be a push-button operated self-closing type capable of delivery of 40 to 60 liters oxygen flow per minute when the delivery pressure measured at the oxygen inlet connection of the operative head is 50 psig. Valve shall be color coded white and marked "OXYGEN FLUSH".

3.6.6 Breathing pressure gage. The breathing pressure gage shall be of the aneroid type, calibrated from -10 mm Hg to +70 mm Hg and -20 centimeters of water (cm H_2O) to +100 cm H_2O. The gage shall have a threaded connection, geared movement, corrosion-resistant metal threaded bezel and case, and beveled plastic window. The maximum error shall not exceed 1-1/2 mm Hg at any scale reading, or 5 percent of indicated reading, whichever is greater.

3.6.7 Adjustable pressure relief valve. One pressure relief valve shall be an adjustable, spring-loaded, non-calibrated type with a range of at least 3 to 60 mm Hg, and shall provide a setting affording positive closure, making the valve inoperative. The full range of adjustment shall require at least two full turns of the adjustment knob.

The relief valve shall vent through a 19 mm male nipple as specified in American National Standards Institute (ANSI) Z79.11. An additional coaxial nipple of 30 mm (specified in ANSI Z79.11) is permissible; no other sized nipples shall be acceptable.

All adjustments of the valve to predictable pressures shall require reference to the breathing pressure gage while the relief valve is venting excess gas.

Relief valve exhaust nipple shall be labeled. Relief valve shall be labeled with instructions to adjust valve to it lowest pressure when not in use.

The pressure required to allow a flow of 1 LPM of oxygen at the lowest pressure adjustment shall not exceed 8 mm Hg and shall drop abruptly to a maximum of 3 mm Hg when the valve is first tested for venting. Immediate repetition of the test shall produce no pressure in excess of 3 mm Hg.

The pressure shall not exceed 33 mm Hg while venting a gas flow mixture of 6 LPM of oxygen and 6 LPM of nitrous oxide after the valve has been initially set to a pressure of 25 mm Hg while venting a flow of 1 LPM of oxygen. Test procedure shall be as specified in 4.4.2.2.

3.6.8 Non-adjustable pressure relief valve. A second pressure relief valve shall be a non-adjustable, spring loaded, uncalibrated type. The valve shall produce a pressure of at least 40 mm Hg when venting a flow rate of 50 cc per minute, and a pressure between 60 and 80 mm Hg when venting a flow rate of 60 LPM. The valve shall vent directly to atmosphere. The valve shall be labeled "NON-ADJUSTABLE".

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3.6.9 Inhalation and exhalation check valves. The inhalation and exhalation check valves shall be of the gravity type, employing a captive disk resting on an angular knife-edged seat. The disks shall be of white, low density polyethylene. The top surface of the valve housing shall be made of a clear suitable material. The flow resistance of each valve to atmospheric air at 30 LPM shall not exceed 0.1 inch water (H_2O). Check valves shall prevent reverse flow in excess of 2 LPM at any differential pressure up to 15 mm Hg.

3.6.10 Utility tray mounting studs. Two mounting studs shall be furnished and assembled to the rear of the operative head. The studs shall provide a suitable mount for the utility tray and in addition shall be capable of accommodating the military mount of National Stock Number (NSN) 6515-00-245-9790 Mounting Kit, Inhaler Unit, Gas Anesthesia Apparatus which, in turn, shall be capable of assembling NSN 6515-00-890-1685 Inhaler Unit, Gas Anesthesia Apparatus, 150 cc capacity to anesthesia apparatus. The studs shall be as shown on the drawing.

3.6.11 Apparatus mount. The apparatus mount shall be capable of supporting the entire operative head assembly. Mount shall be capable of easy unfolding and locking to arms of case when placing the apparatus in the normal operating or storage position. Mount shall be fabricated from forged aluminum alloy in accordance with ASTM B 247 for 2014-T6 and shall be as shown on the drawing.

3.6.12 Level indicating device. The apparatus shall have a line level permanently secured to the right side arm assembly. The method of installing the line level shall be as follows: Position anesthesia apparatus so that the base is exactly level, end to end. Place the line level inside the right side arm assembly so that the window of the level is upright and visible through the bototm of the rectangular cutout of the right side arm stand. Position the line level to indicate the level position with respect to the base. Mark and center-tap front and rear flange of the arm stand at the point in-line with end plate holes of the line level. Drill holes for #6 screw perpendicular to the flange. Mount the line level between the drilled holes using 6-32 screws, nuts, and lock washers. Adjust position of line level by bending end plates until level is indicated with base of unit level and line level secure. Epoxy top and bottom of line level and allow to dry.

3.6.13 Nitrous oxide shut-off valve. The operative head shall be provided with a normally closed shut-off valve in the reduced pressure supply to the nitrous oxide flowmeter needle valve. This valve shall remain closed whenever the oxygen supply pressure to the oxygen flowmeter needle valves is less than 15 psig, and shall be open whenever that pressure is above 25 psig, when tested as specified in 4.4.3.

3.7 Absorber. The absorber assembly shall consist of two identical removable nested transparent canisters clamped together in a gas-tight relation with the underside of the operative head and the top surface of the absorber base. The absorber base shall be fabricated from brass nickel-plated with a brushed satin finish. The absorber base shall be as shown on the drawing. The canisters shall be made of clear rigid polyvinyl chloride. A very pale blue tint shall be permissible. Each canister shall accommodate approximately 90 cubic inches of absorbent, and shall have a removable perforated bottom and open top.

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The canisters shall be enclosed by attached wire guards in electrical contact with the absorber frame when absorber is in use. The absorbent containers shall accommodate prepacked disposable, transparent cartridges of absorbent conforming to requirements for Soda Lime, USP, 2-1/2 lb., NSN 6505-00-782-6484, in place of bulk absorbent at a sacrifice of not more than 15 percent of absorbent volume. A stud with a knurled head shall be assembled through the apparatus mount to provide the clamping force necessary to seal the canisters upon the absorber base. Travel of stud shall be adequate to permit easy installation and removal of each absorbent canister. Alignment of the operative head and base shall be maintained by two guide posts (in accordance with drawing) attached to the sides of the operative head and assembled beneath the absorber base to the apparatus mount supporting the clamping screw. The method of assembly of absorber to operative head and anesthesia mount shall be as shown on the drawing. Guide posts shall be fabricated from corrosion-resistant metal. The base shall include a condensate drain cock that will effect a leakproof seal when it is in the closed position and the apparatus is operating. The flow resistance to 60 LPM of oxygen flowing from the bag connection through the absorber shall not exceed 2.0 inches H_2O with a full packing of fresh 4-8 mesh soda lime and shall not exceed 2.5 inches of H_2O when filled with two fresh disposable type cartridges. Soda lime shall conform to the requirements of NSN 6505-00-153-8515. A statement calling attention to the need for emptying of the canisters prior to repacking of the anesthesia machine within its integral carrying case shall be hot stamped, in red, on the canister bodies.

3.8 Vaporizer. The vaporizer shall consist of a vaporizer head and a removable vaporizing chamber. The vaporizer head shall include inlet and outlet fittings and hand-operated control valve permitting the vaporizer to be turned on and off. The valve shall, when turned on, divert all oxygen received from the vaporizer flowmeters through the vaporizing chamber and when turned off, allow all such oxygen to bypass the vaporizing chamber while simultaneously venting the chamber to atmosphere. The vaporizer head shall be fabricated from forged brass and shall be nickel plated. The head shall support a suitable bubbler and baffle arrangement to pass the vehicle oxygen through the liquid anesthetic agent to substantially saturate it and to remove liquid droplets from the gas stream before delivering the gas and vapor mixture to the breathing circuit. The chamber shall be fabricated from brass and shall be nickel plated. The chamber shall be equipped with a liquid level sight glass, a filling funnel with a captive funnel plug, a dial thermometer of bimetallic construction in intimate contact with the liquid agent, and a drain cock with a captive plug. The liquid capacity of the chamber at maximum operating level shall be at least 300 cc and the maximum safe level shall be indicated. All markings, including a warning to empty the vaporizer prior to repacking the unit, shall be permanent. In addition, a label shall be affixed across the top front of the vaporizer that reads: "ARMY AND AIR FORCE POLICY REQUIRES THE USE OF AN INLINE OXYGEN MONITOR." The label shall not interfere with the ready visibility of other instructions or the liquid level sight glass. Maximum leakage of the vaporizing chamber shall be 20 cc of oxygen at 50 mm Hg when the operative head is tested as specified in 4.4.2.1.

3.8.1 Vaporizer inlet connection. The vaporizer inlet connection shall be permanently secured to the vaporizer head and shall be capable of being assembled to, or disassembled from, the balance of the operative head using one of the open end wrenches furnished.

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3.8.2 Vaporizer outlet connection. The vaporizer outlet connection shall be a removable inhaler tubing adapter. The outlet adapter shall be assembled to the vaporizer head in a secure, leakproof manner. The adapter shall function as a standard 22 mm male tapered connection and shall be gaged as shown in ANSI Z79.8 (gage shall be as shown in figure 2 of the ANSI standard). Adapter shall be fabricated from corrosion-resistant metal. This connection shall be the common gas outlet of the apparatus.

3.8.3 Vaporizer performance. The vaporizer shall be capable of delivering, at any rate of vehicle-oxygen-flow within the range of 20 to 1000 cc per minute, a mixture of oxygen and the vapor of a liquid anesthetic agent which is at least 80% saturated with vapor at the temperature of the vaporizer thermometer, providing that the liquid content of the chamber is within 250 cc of that necessary to achieve the maximum safe level, and that operation is conducted in a quiet, ambient atmosphere with relative humidity low enough to prevent condensation on the vaporizer. Upon request by the Government, the contractor shall be required to furnish test results to demonstrate such performance.

3.9 Oxygen and nitrous oxide pressure regulators. Two oxygen and two nitrous oxide pressure regulators shall be furnished with each anesthesia apparatus assembly. The regulators shall be of the automatic single-stage pressure-reduction type. The regulators shall be identified by a colored label at least 9/16-inch in diameter, white for oxygen and blue for nitrous oxide. The color shall form a background for the chemical symbol for the corresponding gas: "O_2" for oxygen and "N_2O" for nitrous oxide. The style and design of the regulators shall be as shown on the drawing. Dimensions shall be such as to fit the cut-outs provided in the carrying case and designed to be properly assembled to "D" size cylinders without interference with adjacent cylinders and regulators when unit is erected for use. Regulators shall possess integral pressure gage and outlet check valve. The body of the regulator shall be fabricated from forged or machined copper alloy, nickel plated or chromium plated over nickel plated; Monel alloy 400 or Inconel 600. Alloys shall have a minimum tensile strength of 50,000 psi. Polytetrafluoroethylene (Teflon), polychlorotrifluoroethylene (KEL-F), or vinylidenefluoride hexafluoropropylene (Viton) shall be the only acceptable polymeric materials used in the internal construction of the regulators. All other plastic or rubber compounds shall be prohibited.

3.9.1 Pressure regulator performance. Regulators shall be set to deliver a pressure of 45 psig to an orifice of 0.025 inch diameter opening to 1 atmosphere absolute pressure when the gas inlet pressure is 100 psig. Raising the inlet pressure to 2000 psig shall change the delivery pressure by not more than 10 psig. The regulators shall provide a flow of at least 55 LPM of oxygen measured at 1 atmosphere absolute pressure when previously adjusted as specified above, and subsequently allowed to flow through an orifice of 0.060 inch diameter with a gas inlet pressure of 100 psig. The regulators shall have leakage no greater than 100 psi per hour at any gage pressure above 1000 psi.

3.9.2 Regulator inlet. Each regulator shall have a yoke type pin-index integral with the regulator body, in accordance with the following inlet connection numbers of ANSI B57.1: Connection No. 870 for the oxygen regulator, and Connection No. 910 for the nitrous oxide regulator. Each regulator shall have a replaceable stainless steel yoke nipple containing a porous bronze filter, and each regulator valve shall be completely replaceable. The inlet passage shall provide a dead-end chamber to adequately absorb the heat of recompression when

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the inlet is suddenly pressurized to 2000 psig. The yoke clamp screw shall be tee-handled style with 3/8 inch-16 thread. The handle shall be colored white to identify it for oxygen, or blue to identify it for nitrous oxide, and shall be permanently stamped with the chemical symbol for the corresponding gas.

3.9.3 Regulator relief valve. Each regulator shall be furnished with a spring-loaded pressure relief valve set to open at more than 100 psig and less than 200 psig and having sufficient flow capacity to prevent the outlet pressure from exceeding 300 psig under any inlet pressure up to 2000 psig.

3.9.4 Regulator pressure gage. Each regulator shall have a 0-3000 (minimum) psig pressure gage mounted in direct communication with the high-pressure chamber. The maximum error over the middle half of the scale shall not exceed 60 psig. Maximum error over the first and fourth quarters of the scale shall not exceed 90 psig. The gage case shall be corrosion-resistant material, with threaded bezel retaining a bevelled plastic cover.

3.9.5 Regulator outlets. The regulator outlets shall be fitted with automatic check valves, Schrader No. 4951 or equal, for oxygen, and Schrader No. 4826A or equal, for nitrous oxide.

3.9.6 Thread sealants. Organic-based thread sealants which are considered to be flammable when in contact with oxygen shall not be used. Thread sealants used for antiseize and sealing purposes shall be teflon thread-sealing tape or thread compounds in strict acordance with MIL-T-5542.

3.10 Oxygen and nitrous oxide regulator adapters. The oxygen and nitrous oxide regulator adapters shall be provided with appropriate pin-indexing, inlet nuts, glands, and flush type outlet connections in accordance with the following connection numbers of ANSI B57.1. Connection numbers 540 and 870 for the oxygen regulator adapter, and connection numbers 326 and 910 for the nitrous oxide regulator adapter. The adapter bodies shall be of copper alloy, with the material restrictions specified for the regulators. The adapters shall support the regulator in the same orientation when attached to a screw connection cylinder as the regulator would have when attached without its adapter to a flush type cylinder. Adapters shall be capable of being attached to, or removed from, cylinders using accessory open end wrenches. The style, design, and dimensions of the adapters shall as shown on the drawing. No leakage of the adapters shall be permissible at 2000 psig.

3.11 Flexible hose assemblies. Oxygen and nitrous oxide hose assemblies shall be electrically conductive from end to end; and hose (less connections) shall be listed by the UL in Classification 400 UO. Hose shall be constructed of two plies of rubber enclosing one ply of fabric, and shall have minimum nominal working pressure of 250 psig. Hose shall have a nominal I.D. of 1/4-inch and a nominal O.D. of 1/2-inch. Hose shall be all black (not color coded) US Rubber Co., "USR Conductive" hose, or equal. The hoses shall be as shown on the drawing. All hoses shall be measured while a tensile load of 5 pounds maximum is applied.

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3.11.1 Flexible hose fittings. The oxygen and nitrous oxide flexible pressure hose assemblies shall each include an inlet adapter and an automatic outlet check valve. The inlet adapter for the oxygen hoses shall be Schrader No. 4927-198 or equal; for nitrous oxide hoses, shall be Schrader No. 4934-198 or equal. The automatic outlet check valve for oxygen hoses shall be Schrader No. 4951 or equal; for nitrous oxide hoses, shall be Schrader No. 4826A or equal. No leakage of the oxygen and nitrous oxide hose assemblies shall be permissible with internal pressure of 100 psig. Fittings at the end of each hose shall be permanently and legibly color coded; white for oxygen and blue for nitrous oxide.

3.12 Rubber goods. Each anesthesia apparatus shall be furnished with a head harness, two rebreathing bags, five oronasal masks, three inhaler tubings, and one supply tubing.

3.12.1 Head harness. The head harness shall be one piece, conductive rubber and shall have four straps perforated at intervals which shall appropriately fasten to the hook rings of the three larger oronasal masks. The head harness shall be in accordance with the requirements specified for NSN 6515-00-386-9708.

3.12.2 Rebreathing bags. The rebreathing bags shall be fabricated from conductive rubber. The adult rebreathing bag shall have a capacity of 3 liters. The opening of the adult rebreathing bag shall meet the requirements of Compressed Gas Association's (CGA) Pamphlet M-1 for 22 mm male connectors and suitably fit over the bag connector on the apparatus. The pediatric rebreathing bag shall have a capacity of 1 liter. The opening of the pediatric bag shall have a metallic bushing that shall accept 15 mm endotracheal tube connectors and adapters that are in accordance with ANSI Z79.2. The opening of the pediatric bag shall be gaged in accordance with ANSI Z79.2. The opposite end of the pediatric bag shall have a nipple containing an adjustable bleed valve and a fitting for connection to an inhaler tube or similar component for delivery of excess gas to a remote location.

3.12.3 Oronasal masks. The oronasal masks shall be fabricated from conductive rubber. The large and medium adult size anatomical masks shall meet the requirements specified for NSNs 6515-00-299-8297 and 6515-00-347-2200, large adult and medium adult, respectively. The newborn, pediatric, and child size Rendell-Baker-Soucek masks shall be in accordance with the requirements specified for NSNs 6515-01-033-3693, 6515-01-033-3694, and 6515-01-033-3695, respectively.

3.12.4 Inhaler tubings. The two 32-inch long inhaler tubings shall be fabricated from corrugated conductive rubber. These inhaler tubings shall provide flexible conduits which shall connect the vaporizer and operative head to the "Y"-connector. Each tubing shall have a rubber bushing in both ends. The tubings shall meet the requirements for NSN 6515-00-302-1500. The one 10-inch long inhaler tubing shall be fabricated from corrugated conductive rubber to provide a flexible conduit between the pediatric rebreathing bag and a Tee-connector. One end of the tubing shall have a bushing with a standard 15 mm male connector in accordance with ANSI Z79.2. The other end of the tubing shall have a bushing with an internal bore to suitably fit the 15 mm male connector.

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3.12.5 "Y"-connector. The "Y"-connector shall be a one-piece fitting of three nipples fabricated of rigid conductive reinforced thermoplastic material. Two of the nipples shall have their axis divergent by approximately 35°, and shall have 22 mm male tapered connections in accordance with CGA Pamphlet M-1 and shall be gaged with the ring gage specified in ANSI Z79.8 (Figure 2). The axis of the third nipple shall bisect the angle between the nipples (male) and shall have an internal bore that is a standard 15 mm female connector/adapter in accordance with ANSI Z79.2. The 15 mm female connection shall be gaged in accordance with ANSI Z79.2.

3.13 Protective closures. The following openings shall be protected with closure devices to prevent foreign material, such as dust, dirt, chips, flakes, etc., from entering the apparatus:

Regulator outlet - (4 closures required)
Cylinder adapter outlet - (2 closures required)
Gas inlet connections - (2 closures required)
Vaporizer outlet (common gas outlet) - (1 closure required)
Exhalation inlet - (1 closure required)
Rebreathing bag connection - (1 closure required)

All closures shall be of the captive type. The closures for the regulator outlets shall be Schrader glands with rubber caps. The Schrader glands shall be of the blind type and shall lock into the corresponding Schrader regulator outlet. The closures for the two cylinder adapters shall be of the plastic-cap type, or equivalent, that shall suitably seal the adapter outlet.

The closures for the gas inlet connections shall be of the rubber-cap type, or equivalent, that shall suitably seal the gas inlet connections.

The remaining outlets and inlets (vaporizer, rebreathing bag connection and exhalation inlet) shall be of the rubber-stopper type, or equivalent, that shall suitably seal the openings.

3.14 Accessories. In addition to the number and type of anesthesia component parts previously specified, the following additional accessory items shall be provided:

a. Two open-end wrenches of corrosion-resistant, nonferrous, nonsparking metal one wrench, shall have an opening for attaching regulator adapter (see 3.10) to large cylinders on one end, and an opening for the vaporizer connection (see 3.8.2) at the other end; the other wrench shall have an opening for use on the needle valve packing nuts at one end, and an opening for the tray mounting studs at the other end. Wrenches shall be as shown on the drawing.

b. A 3/16-inch L-type Allen wrench of corrosion-resistant, nonferrous, nonsparking metal.

c. A tee-handle wrench for operation of the gas cylinder valves. The style and design of the wrench shall be as shown on the drawing.

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d. An endotracheal elbow adapter having a 15 mm male taper, in accordance with ANSI Z79.2 at one end and a 22 mm male external tapered connection (gaged as shown in ANSI Z79.8) and coaxial 15 mm tapered female bore at the other end. The 15 mm tapered female bore shall be gaged in accordance with ANSI Z79.2. Adapter shall be black Sanford processed aluminum. Shall be as shown on the drawing.

e. A utility tray capable of assembly to the operative head mounting studs and supported by a steel rod support. Tray shall be as shown on the drawing.

f. A clipboard capable of being supported on the utility tray or mounted in cut-outs provided in the pedestal arms. Shall be as shown on the drawing.

g. A corrosion-resistant metallic adapter with one end sized to suitably fit into the opening of the vaporizer outlet connection (common gas outlet) - this end shall possess an O-ring to form a leakproof junction; the other end of the adapter shall be sized to form a leakproof junction with 1/4 inch I.D. rubber supply tubing.

h. A fifty-inch length of 1/4 inch I.D., 1/16 inch wall, conductive rubber supply tubing.

i. A tee-adapter having one 15 mm male and one 15 mm female connector, all in accordance with ANSI Z79.2 (and gaged in accordance with the standard), on a common axis and a male nipple at a right-angle to that axis to fit 1/4 inch I.D. rubber supply tubing. The adapter shall be fabricated from corrosion-resistant, conductive material.

3.14.1 Repair parts. The following repair parts shall be furnished with the anesthesia apparatus:

10 each cylinder valve sealing washers for small oxygen and nitrous oxide cylinders. Only the polymeric materials specified for the oxygen regulators (see 3.9) shall be acceptable.
2 each discs for inhalation and exhalation check valves.
1 each absorber canister gasket.
1 each vaporizer funnel plug with captive chain and clip.
1 each vaporizer drain plug with captive chain and clip.
2 each valve seats, high pressure side for oxygen and nitrous oxide pressure regulators. The valve seats shall be fabricated from the polymeric materials specified for the oxygen valves (see 3.9) or austenitic (300 series) corrosion-resistant steel.

All repair parts shall be suitably retained and protected within the carrying case, and referenced in the operation and repair manual together with appropriate installation instructions and identification by manufacturer's part number and description.

3.15 Carrying case. The carrying case shall consist of a body and cover with integral relief valve. The body shall house the operative head assembly in the retracted position, and shall include a folding means for supporting and locking the head at normal working height when the apparatus is erected for use.

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In addition, the body shall provide storage facilities for conductive casters, sockets for receiving slides or casters, mounting for the regulator adapters, face masks, wrenches, a utility tray, a cylinder support rack, and a compartment for storage of small accessories. The cover of the case shall be completely removable from the body, and shall be ruggedly constructed to provide a seat for the operator. Case cover shall be fitted with a neoprene, non-conductive gasket fabricated as shown on the drawing to provide a rain-proof closure. Cover shall provide storage space and securing means for flexible hoses, breathing tubes, regulators, rebreathing bags, a clip board and a mask retainer. Case shall be of aluminum alloy construction and shall have smooth, rounded edges and corners on exterior surfaces. Cover shall be secured by means of four adjustable toggle type drawbolts. Drawbolts shall be adjusted and locked in place in such a manner that there is a positive compression applied on the case gasket (but not an excessive amount that will cause the walls of the case to buckle). Body shall have one folding bail-type handle on each end. The case construction shall be in strict accordance with drawing.

3.15.1 Case relief valve. The case relief valve shall have a nominal relief pressure rating of 1.5 psig in both directions at an air flow of 20 SCFM and a nominal reset pressure of 1.0 psig. The relief valve shall be tested as follows: Remove cover of case and replace with text fixture sealed to case body; the pressure rise shll not exceed 2.5 psig at an air flow of 55 to 65 LPM. The case pressure shall fall to not less than 1.0 psig when the air flow is reduced to zero. With the test fixture removed and the exterior portion of relief valve covered with a second test fixture sealed against the case body, pressure shall be not more than 2.5 psig when an air flow of 55 to 65 LPM is delivered to the test fixture. Pressure in the fixture shall not drop to less than 1.0 psig when the air flow is reduced to zero.

3.15.2 Sealing. All spot-welded seams of the case shall be sealed prior to spot welding to produce a waterproof joint. (Sealing such as Minnesota Mining and Manufacturing Company's adhesive EC-2214, or its equal, is acceptable.) Aluminum surfaces shall be thoroughly cleaned, dry, and grease-free prior to application of adhesive. Cleaning shall be by means of vapor degreasing, alkaline degreasing or acid etch. Care shall be taken not to incorporate air into the adhesive during application. Surfaces shall be spot welded and excess adhesive cleaned prior to curing with a ketone-type solvent. Adhesive bond shall be cured by a time-temperature bake following the recommendations of the adhesive manufacturer. The finished case, including the gasket, shall be rain-proof when tested as specified in 4.4.5 in that there shall be no entry of water into the case.

3.15.3 Finish. All accessible aluminum alloy surfaces of the case and head support, both inside and outside, shall be finished with enamel as specified herein. Surfaces to be enameled shall be thoroughly cleaned and degreased and then shall receive a zinc chromate primer conforming to TT-P-1757, which shall be followed by two coats of lusterless olive drab enamel conforming to TT-E-527 and to color 34087 of FED-STD-595. All enamel coats shall be sprayed on to accomplish a complete and uniform coverage of the coated parts, and shall be baked to give a smooth, hard and tough finish, that shall be free from runs and resist cracking and chipping. Contractor shall endeavor to control film thickness so that thickness of a single dried coat is 0.25 to 0.35 mils for primer and 1.0 to 1.5 mils for enamel.

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3.15.4 Casters and glides. Each unit shall be provided with four stem type steel, cadmium-plated glides, having a base diameter of 1-1/2 inches. In addition, four electrically conductive casters shall be stored in the caster retainer in the case of the anesthesia machine. The casters shall conform to FF-C-77, Type II, Class A, Style b, Mounting N, no less than 3 inches in diameter with 100-pound minimum capacity for each wheel. (Reference: Bassick Co. part no. H13695TX2.) The average electrical resistance of each caster shall not exceed 250,000 ohms and the maximum shall not exceed 1,000,000 ohms when tested as specified in 4.4.1.1. Casters and glides shall have stems fitted with friction rings to securely engage in the sockets specified in the drawing.

3.16 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. This information shall specifically include instructions and drawings. The service data shall include such detailed information as will permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Drawings and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. The service manual shall contain the following precautionary note:

> "CAUTION - DO NOT USE ORGANIC-BASED THREAD SEALANTS ON ANY PORTION OF THE OXYGEN REGULATOR. USE ONLY TEFLON THREAD-SEALING TAPE OR THREAD COMPOUND, ANTISEIZE AND SEALING, OXYGEN SYSTEMS, 4 OZ. TUBE, NSN 8030-00-243-3284, AVAILABLE FROM THE GENERAL SERVICES ADMINISTRATION."

As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope, shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, the contents of the exterior envelope may be limited to information necessary for roughing-in preparations, and a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located.

3.17 Identification marking. The apparatus carrying case shall have a metal plate permanently affixed thereto. The plate shall bear, in addition to any other markings required, the name and address of the manufacturer (if the

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manufacturer is not the contractor, the contractor's name and address shall also appear), and the NSN of the item. The markings shall be clear and permanent. Model and serial number of the apparatus shall be marked on the apparatus.

3.17.1 Instruction marking. On the insert cover there shall be legibly and permanently marked information covering the erection and repacking instructions for the anesthesia apparatus. A draft of the instructions shall be submitted to the contracting officer for approval. Marking may be engraved directly on the insert cover or on a metal plate permanently affixed to this cover, and cover spotwelded or riveted in place. Instructions shall be clearly visible once the carrying case cover is removed from the body.

3.18 Welding. Welding shall be done by qualified welders using appropriate equipment in a qualified process. Spot-welds shall be capable of developing the tensile strength of the parent metal. All exterior spot-welds on the case and uprights shall be sufficiently smoothed so as not to be visible after the enamel finish is applied.

3.19 Model number. In order to maintain continuity of various field anesthesia apparatus procured by the military medical services, the model number associated with the design specified herein shall be designated "MODEL 885A". The model number shall appear on the flowmeter manifold as previously specified herein, and on the cover of the instruction and service manual.

3.19.1 Acceptable variations. Variations to the design specified shall be acceptable, provided that end use of item and interchangeability requirements and physical aspects are not affected. The acceptance of the variations shall be granted only by the contracting officer.

3.20 Tooling and gages. All tooling required to manufacture the entire anesthesia apparatus shall be the sole responsibility of the contractor. The Government shall assume no responsibility whatsoever in furnishing any tooling nor be required to procure any devices beyond that specifically contracted in any contract. All gages to inspect the 15 mm and 22 mm male and female adapters, connections, etc., specified herein shall be furnished by the contractor and made available to the Government representative for required testing of such components.

3.21 Vaporizer. The vaporizer specified herein shall afford the capability of utilizing current and volatile anesthetic agents without the use of individual specialized vaporizers.

3.22 Contractor certification. All rubber goods that are required to be electrically conductive and are manufactured as commercial products by the contractor, or otherwise manufactured commercially, shall be certified by the contractor that the rubber components meet the salient characteristics as specified herein and conform to the producers' own drawings, specifications, standards, and quality assurance practices, and is the same as the product or component sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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3.23 Workmanship. The anesthesia apparatus shall be free from defects which affect the appearance or impair the serviceability of the apparatus.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	II	2.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for performance test shall be as indicated for inspection level I, AQL 0.65 in MIL-STD-105.

4.2.2.2 Sampling for other tests. Sample size for each test shall be as indicated for inspection level S-1 in MIL-STD-105, except that the acceptance number shall be zero for all sample sizes, and a minimum of two samples shall be taken regardless of lot size.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination. The anesthesia apparatus shall be examined to determine compliance with specification requirements. Examination shall be conducted as specified in tables I and II. Inspection, however, shall not be restricted to defects listed in table II. Any anesthesia units in the sample containing one or more defects shall be rejected and, if the number of defective items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE II. Classification of defects.

Categories	Defects
Major	
101	All external brass components not plated.
102	One or more components missing (including accessories and repair parts).
103	All rubber goods not marked "conductive" or "electrically conductive", such as: oxygen and nitrous oxide flexible hoses, rebreathing bags, oronasal masks, inhaler tubings, head harness, and supply tubing.
104	Apparatus not of the design specified.
105	Operative head and inlet connections not marked with the symbols "O_2" and "N_2O" adjacent or upon the inlet connections.
106	Flowmeter scales, needle valves, oxygen flush valve, oxygen pressure regulator, and nitrous oxide pressure regulators not color coded as specified.
107	Flowmeter scale graduations not as specified.
108	Flowmeter manifold does not indicate markings as specified.
109	Breathing pressure gage scales not as specified.
110	Adjustable and non-adjustable relief valves not labeled as specified.
111	Vaporizer inlet connection and vaporizer outlet connection not capable of easy assembly to or from the operative head.
112	Regulator inlets and regulator adapters not pin-indexed as specified.
113	Regulator outlets not of the "quick disconnect" (Schrader) design.
114	Oxygen and nitrous pressure hoses do not possess fittings as specified.
115	Protective closures missing where specified.
116	Service data not furnished.
117	Apparatus not free from sharp edges, burrs, and slivers on exposed surfaces (including carrying case).
118	Components not free from corrosion, scale, pits, holes, and crevices.
119	Carrying case closure not as specified.
120	Absorber canisters not in electrical (conductive) contact with absorber frame.
121	Certification of UL approval (or approved laboratory) missing on appropriate components.
122	Model number and serial number missing.
Minor	
201	Carrying case not identified as specified.
202	Insert cover marking not as specified.
203	Regulator pressure gage face not as specified.

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4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Electrical conductivity. Samples which have been subjected to 100 percent elongation shall not be used for this test. The samples shall be conditioned in air at ordinary room temperature, with relative humidity not greater than 80 percent, for not less than 24 hours; then shall be tested under the same ambient conditions. The resistance between the electrodes shall be measured by any resistance measuring apparatus which has an open-circuit direct voltage of approximately 500 volts. (For safety of the operator, the maximum current which the measuring device is capable of delivering through a resistance of 500 ohms should be less than 10 milliamperes.)

4.4.1.1 Casters. Electrically conductive casters shall be tested for electrical resistance while supporting the anesthesia apparatus, or they shall be tested while supporting 25 percent of the rated load. The surface of the tire shall be clean and dry; it may be cleaned by any method that removes wax and dirt, but does not abrade the tire surface. The caster, after being placed under the load, shall be rolled on a clean, dry, flat metal plate. The metal plate and the metal parts of the caster shall be used as electrodes for the test. If tested while supporting equipment, all other casters under the equipment shall be insulated from the floor on which the plate is placed. Resistance measurements shall be made with five separate areas of the tire in contact with the plate, and the average and maximum resistance determined.

4.4.1.2 Face mask connection. The electrical resistance between the metal parts of the apparatus and the face mask, when properly connected, shall be measured. One electrode shall be connected to a metallic plate, 6 inches by 6 inches, against which the face mask shall be pressed by a 5-pound load.

4.4.2 Leakage tests.

4.4.2.1 For complete operative head assembly. The adjustable relief valve shall be closed. The rebreathing bag connector shall be plugged in a leakproof manner. An external segment of suitable tubing shall be connected to the outlet of the vaporizer and the exhalation check valve connector in a leakproof manner. Oxygen, at a maximum flow rate of 200 cc per minute, shall be internally admitted to the operative head with the vaporizer control valve set in the open position. The internal gage pressure reading, as indicated by the breathing pressure gage, shall be at least 50 mm Hg. The test shall be repeated with the vaporizer control valve set in the closed position and the same procedure specified above shall be followed. The reading of breathing pressure gage shall, again, be at least 50 mm Hg.

4.4.2.2 Adjustable pressure relief valve. One end of a suitable inhaler tube shall be connected to the vaporzier outlet, and the other end connected to the exhalation valve inlet. The rebreathing bag connector shall be plugged in a leakproof manner. Oxygen and nitrous oxide shall each be supplied to the inlet connectors at a minimum pressure of 30 psig with all needle valves initially closed.

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The adjustment knob of the relief valve shall be turned counterclockwise to the end of its travel. The breathing pressure gage shall read within 1 mm Hg of zero. Then the oxygen flowmeter needle valve shall be slowly opened to establish an oxygen flow of 1 LPM. The pressure as indicated by the breathing pressure gage shall increase by not more than 8 mm Hg in the next ten seconds and then fall abruptly to not more than 3 mm Hg. Immediate repetition of this test shall produce no pressure greater than 3 mm Hg.

With the oxygen flow still adjusted to 1 LPM, readjust the relief valve to raise the breathing pressure gage to 25 mm Hg. Then adjust the flow of oxygen and nitrous oxide to 6 LPM each. The circuit pressure gage shall then read not more than 33 mm Hg.

Adjust the relief valve slowly to increase the pressure at the breathing pressure gage. When the pressure reaches 60 mm Hg, verify that there is at least 30° of adjustment remaining before the clockwise limit of travel is reached. Then readjust the valve knob to the counterclockwise limit of its travel, and verify that at least two full turns are required.

4.4.3 Nitrous oxide shut-off valve performance. The operative head shall be provided with a supply of nitrous oxide at a pressure of 40 psig, and the nitrous oxide flowmeter needle valve opened one full turn. There shall be no flow indicated on the nitrous oxide flowmeter. The operative head shall then be connected to an oxygen supply at a pressure of 10 psig or less, without affecting the nitrous oxide flowmeter. The oxygen supply pressure shall then be increased slowly. Flow, as indicated by the nitrous oxide flowmeter, shall appear before the oxygen pressure exceeds 25 psig. The oxygen supply pressure shall then be reduced. Nitrous oxide flow shall cease at, or above, an oxygen supply pressure of 15 psig.

4.4.4 Flowmeter accuracy. Only oxygen and nitrous oxide calibrated flowmeters with accuracy traceable to the National Bureau of Standards (NBS) shall be used. The oxygen calibrated test flowmeter shall have a flow range of 20 cc/minute to 7 LPM, and the nitrous oxide calibrated test flowmeter shall have a flow range of 200 cc/minute to 8 LPM.

Connect inlet of the appropriate calibrated test flowmeter to the outlet of the vaporizer. Adjust the appropriate flowmeter needle valve (flowmeter being tested) to each value on the flowmeter scale of the apparatus. The flow rates indicated on the apparatus flowmeter shall be compared with the flow indicated on the calibrated test flowmeters with the appropriate gas entering the corresponding inlet of the operative head.

NOTE: Care should be taken that the calibrated test flowmeters are in calibration during periods of prolonged use.

4.4.5 Rain test. The test item shall be placed in rain chamber and exposed to a simulated rainfall of 4 inches per hour for a total of 30 minutes consisting of 5 minutes on each of its major surfaces. The rainfall shall be dispersed uniformly over the test area, droplets having a minimum diameter of 1.5 mm. Temperature of water shall be between 11°C and 35°C. At the conclusion of test period, the item shall be removed from test chamber and the case examined for entry of water.

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5. PACKAGING

5.1 Packaging.

5.1.1 Unit of issue. One anesthesia apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A. Each anesthesia apparatus with accessories shall be positioned within the carrying case. Positioning of components and closure of the case shall be as specified in 3.14.

5.1.3 Level C. Each unit shall be packaged as specified in 5.1.2.

5.2 Packing.

5.2.1 Level A. Each unit shall be packed in a box designed for a type 1 load and constructed in accordance with PPP-B-601, class 2 or PPP-B-621, overseas type. The box shall be lined with a waterproof case liner conforming to MIL-L-10547. The carrying case shall be cushioned with fiberboard pads or other suitable material to prevent rupture of the case liner. Closure of the box and case liner shall be as specified in the applicable box and case liner specification.

5.2.2 Level C. Each unit shall be packed in a commercial container (export when applicable) so constructed as to insure acceptance and safe delivery by common or other carriers to point of delivery called for in the contract or purchase order.

5.3 Marking. Marking shall be as specified in MIL-STD-129. Marking shall include arrows as specified in 5.4.2 of MIL-STD-129. The legend "FRAGILE" - "HANDLE WITH CARE" shall also be shown.

6. NOTES

6.1 Intended use. This portable anesthesia apparatus is intended for the administration of inhalation anesthesia for use in the field.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers NSN 6515-01-003-4133 ANESTHESIA APPARATUS, GAS, NITROUS OXIDE, appearing in the Federal Supply Catalog.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DoD - MB

Agent:
DLA - DM

Project No. 6515-2586

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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